

المعدل : (10)
المدة : ساعة ونصف

Treat the following exercises:

I- (5 pts)

a) Write the DTD associated to the following XML file:

```
<?xml version="1.0" encoding="ISO-8859-1" ?>
<!DOCTYPE books SYSTEM "books.dtd">
<books>
  <book>
    <number>123</number>
    <title>Les Trois Mousquetaires</title>
    <author>A.Dumas</author>
    <publisher>Gallimard</publisher>
    <price>46€</price>
  </book>
  <book>
    <number>124</number>
    <title>Les Diaboliques</title>
    <author>Boileau</author>
    <author>Narcejac</author>
    <publisher>Duchnok</publisher>
    <price/>
  </book>
</books>
```

b) Display, by using a XSLT file, the data of the XML file in a table.

II- (3 pts)

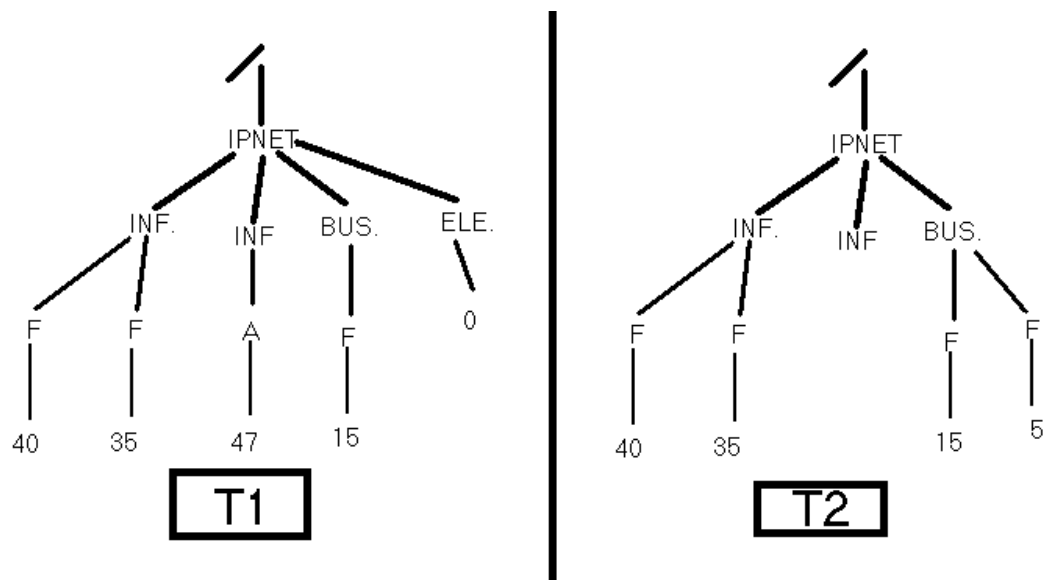
Given the following DTD:

```
<!ELEMENT image EMPTY>
<!ATTLIST image
alt CDATA #IMPLIED
src CDATA #REQUIRED
type CDATA "image/jpg">
```

Among the four XML instances below, what are those that are valid for this DTD?

1. <image alt="birthday" src="birthd.jpg">My birth day </image>
2. <image src="birthd.gif"/>
3. <image alt="birthday"/>
4. <image type="image/gif" src="birthd.gif" alt="birthday"/>

III- (4 pts)



- a) Translate these trees into XML documents X1.xml and X2.xml.
- b) Define a DTD that accepts the two XML documents (trees).
- c) Define a DTD that accepts the first XML document, but not the second.

IV- (4 pts)

Write a XML schema for a bibliography.

- This bibliography contains books and articles ;
- The necessary information for a book (element **book**) are:
 - o its general title (element **title**) ;
 - o the author names (element **author**) ;
 - o its parts (element **part**) and for each part, the number of pages (element **page**) ;
 - o general information on its publishing (element **infoPublishing**) as for example the publisher name (element **publisher**), the publishing place (element **placePublishing**), the printing place (element **placePrinting**), its ISBN number (element **ISBN**) ;
- The necessary information for an article (element **article**) are:
 - o Its title (element **title**) ;
 - o the author names (element **author**) ;
 - o its publication references (element **infoPublication**) : journal name (element **journalName**), pages numbers (element **page**), publication year (element **yearPublication**) and journal number (element **journalNumber**)
- We will reserve also an optional field, for each book and each article, for a personal comment (element **comment**).

V- (4 pts)

Given the following XML document:

```
<?xml version="1.0" encoding="ISO-8859-1"?>

<text>
  <header>
    <title>Advice from an instructor</title>
    <title lang="french">Conseil d'un enseignant</title>
  </header>
  <phrase loc="narrator">
    <translation>The student must be present 5 minutes before the lecture beginning.</translation>
    <translation lang="french">L'étudiant doit se présenter 5 minutes avant le début du cours.</translation>
  </phrase>
  <phrase loc="narrator">
    <translation>The student has never respected the instructor advice</translation>
    <translation lang="french">L'étudiant n'a jamais respecté le conseil de l'enseignant.</translation>
  </phrase>
  <phrase loc="student">
    <translation>But Why? said the student.</translation>
    <translation lang="french">Mais pourquoi? dit l'étudiant.</translation>
  </phrase>
</text>
```

Give the XPath expressions to find in the document:

- a) All the translations
- b) The translations of the 2nd phrase
- c) The translation in English of the 2nd phrase
- d) All the English translations of phrases said by the student
- e) Find all the phrases said by the student and that contain the character '?' (only English translation)